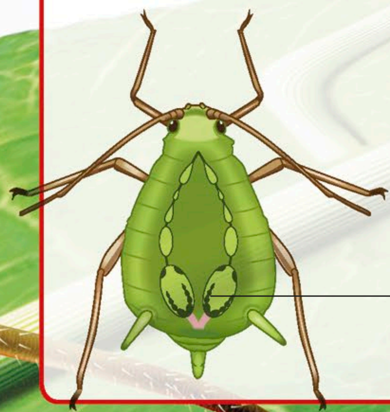
Birth
A female aphid gives birth to live young.

Reproduction

By producing offspring, organisms ensure that their populations survive, as new babies replace the individuals that die. Breeding for most kinds of organisms involves two parents reproducing sexually by producing sex cells. But some organisms can breed asexually from just one parent.

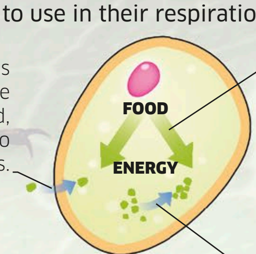


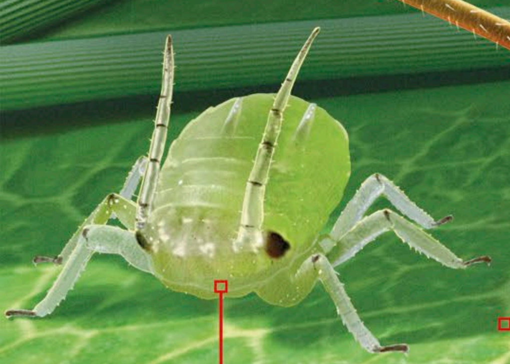
Babies in babies
Some female aphids carry out a form of asexual reproduction where babies develop from unfertilized eggs inside the mother's body. A further generation of babies can develop inside the unborn aphids.

The daughters that are old enough to be born already contain the aphid's granddaughters.

Respiration

Organisms need energy to power their vital functions, such as growth and movement. A chemical process happens inside their cells to release energy, called respiration. It breaks down certain kinds of foods, such as sugar. Most organisms take in oxygen from the environment to use in their respiration.



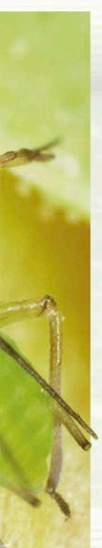


Excretion

Hundreds of chemical reactions happen inside living cells, and many of these reactions produce waste substances that would cause harm if they built up. Excretion is the way an organism gets rid of this waste. Animals have excretory organs, such as kidneys, to remove waste, but plants use their leaves for excretion.

Excretion by leaf
Plant leaves have pores, called stomata, for releasing waste gases, such as oxygen and carbon dioxide.

Growth



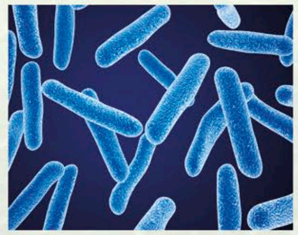
All organisms get bigger as they get older and grow. Single cells grow very slightly and stay microscopic, but many organisms, such as animals and plants, have bodies made up of many interacting cells. As they grow, these cells divide to produce more cells, making the body bigger.

Molting
The body of an aphid is covered in a tough outer skin called an exoskeleton. In order to grow, an aphid must periodically shed this skin so its body can get bigger. Its new skin is initially soft and flexible, but soon toughens.

SEVEN KINGDOMS OF LIFE

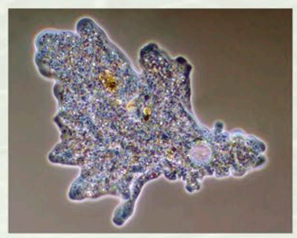
Living things are classified into seven main groups called kingdoms. Each kingdom contains a set of organisms that have evolved to perform the characteristics of life in their own way.

Archaea
Looking similar to bacteria, many of these single-celled organisms survive in very extreme environments, such as hot, acidic pools.



Bacteria
The most abundant organisms on Earth, bacteria are usually single-celled. They either consume food, like animals do, or make it, like plants do.

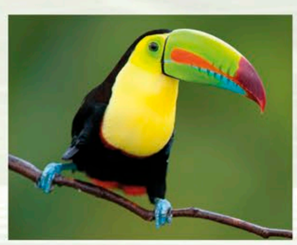
Algae
Simple relatives of plants, algae make food by photosynthesis. Some are single-celled, but others, such as seaweeds and this *Pandorina*, are multicelled.



Protozoa
These single-celled organisms are bigger than bacteria. Many of them behave like miniature animals, by eating other microscopic organisms.

Plants
Most plants are anchored to the ground by roots and have leafy shoots to make food by photosynthesis.

Fungi
This kingdom includes toadstools, mushrooms, and yeasts. They absorb food from their surroundings, often by breaking down dead matter.



Animals
From microscopic worms to giant whales, all animals have bodies made up of large numbers of cells and feed by eating or absorbing food.